



Arlo Heinrich Steyn

Data Scientist and ML Engineer

Contact

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Location

Western Cape, South Africa

Portfolio Website

<https://portfolio-website-arlo-steyn-v1.vercel.app/>

Check it out!

Hobbies

- Exercise
- Reading
- Braai-ing
- Learning
- Sports

Language

English

Afrikaans

Work Experience

Artefact

Sep 2026 - Current

Senior Machine Learning Engineer

Full Stack Data Specialist

Unifi Africa

Jan 2026 - Sep 2026

Data Scientist - Credit Risk

Full Stack Data Specialist

- Developed and productionised end-to-end credit risk scorecards, including behavioural scoring models that determine client risk categories and scores
- Developed and deployed collections prioritisation models across all five country portfolios
- Chair of Credit and Employer Risk Committees, contributing to credit policy, risk decisions, and limit governance.
- Architected and refactored the organisation's GCP feature store, and established a centralised model repository with version-controlled pipelines integrated into the production scoring engine
- Contributed to the development and productionisation of post-scorecard limit proposals
- Work natively in GCP across BigQuery, Dataform and Data Studio

Education

Stellenbosch University Master Of Engineering

2026 - Current

MEng(Industrial) - Struct Data Science - Fulltime

Stellenbosch University Bachelor of Data Science

2021 - 2025

(BDatSci) with a focal area in Computer Science

Midstream College

Matric - 2020

IEB Matriculant

6 Distinctions, 84% Average

The Code Academy

Completed 2023

Full Stack Engineer

Professional Certification: Full-Stack Web Dev

About Me

For more information about me, my career and in-depth descriptions of main projects (plus other projects), feel free to visit my hosted portfolio website:

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Featured Projects

Below is a selection of my own personal and university projects.

Credit Risk Modelling

- Data Science
- Feature Engineering
- LightGBM
- BigQuery
- GCP
- Date: 2026

Data Science

- Data Modelling
- LightGBM
- GCP
- Date: 2026

ML Engineering

- Dataform
- BigQuery
- Bitbucket
- Date: 2026

ML Engineering

- Thesis Project
- CI/CD | Apache
Airflow | Azure
Devops
- Date: 2025

NLP, Data Science

- Data Science Project
- NLP | LLM Fine-Tuning |
Optimization
- Date: 2024

Kenya & Uganda Behavioural Credit Scorecard

Developed and productionised behavioural credit risk scorecards for the Kenyan and Ugandan lending portfolios using LightGBM, achieving AUC lifts of 5 and 6 points respectively over incumbent models. Engineered novel features and improved model freshness, with the new features becoming key predictors across both models. Model outputs directly inform client credit limits and interest rates. Presented both models and corresponding limit proposals to Credit Committees, securing business sign-off and contributing to increased limits and new business opportunities. Updated the organisation's credit change tracker and model health monitoring to support ongoing model performance oversight. The Kenyan scorecard has delivered the lowest 2@3 vintage performance the business has recorded.

Collections Prioritisation Models

Developed and deployed 20+ collections prioritisation models across all five countries in the business, covering approximately 5 million rows of data. Models rank debtors by their likelihood of making a qualifying payment following a call, using customer recency to segment priority. Each model returns a descending priority-ranked call queue, enabling call centres to focus on customers with the highest expected payment likelihood. Models were benchmarked across recency levels to assess lift, with consistently high and stable AUCs across all segments.

Feature Store & Scoring Engine Refactor

Redesigned and rebuilt the company's feature store from scratch in a new GCP project, enforcing standardised naming conventions, removing redundant features, and enabling cross-country feature reusability. Centralised all model assets and code into a dedicated Bitbucket repository, establishing a single source of truth for model versioning and governance across the business. Standardised model inputs and outputs so the scoring engine calls a single unified function regardless of the model, allowing data scientists to easily add and locally test models while the production scoring engine pulls models directly from the repository for deployment and scoring.

Self Healing ML Pipeline

Led a 3-person team to develop an autonomous self-healing machine learning pipeline for retail demand forecasting using Dunnhumby's dataset. The system integrated LSTM, XGBoost, and Random Forest models with ensemble learning, anomaly detection (LSTM autoencoders, drift tests, FGSM), and explainable AI (SHAP, LIME). It featured automated retraining, synthetic data generation, and robust MLOps integration using MLflow, Apache Airflow, and Azure DevOps. Key outcomes included improved forecast accuracy, reduced manual intervention, and scalable CI/CD deployment for real-world retail environments.

Fine-Tuning of a 3B Parameter Model

Fine-tuned Meta's LLaMA-3.2-3B model on 39,668 Q&A pairs from Cross Validated to create a domain-specific AI assistant for machine learning and data science queries. Used LoRA for parameter-efficient training and UnSloth for optimized performance on limited hardware (Kaggle GPUs), with 4-bit quantization and gradient accumulation. Developed two models—one for breadth and another for depth—demonstrating quality-over-quantity advantages. Led ethical web scraping, advanced data wrangling, and transformer-based fine-tuning to build accurate, resource-efficient NLP systems.